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THE EXTENT OF GREATER INDIA, II. PALAEOMAGNETIC DATA FROM THE LADAKH INTRUSIVES AT KARGIL, NORTHWESTERN HIMALAYAS

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The Ladakh Intrusives at Kargil (central Ladakh, northwestern Himalayas), immediately north of the Indus-Tsangpo suture zone, have been studied palaeomagnetically. These intrusives were emplaced before and during the early Tertiary with final closure of the isotopic systems between 49 and 45 m.y. ago.

Six different magnetization components have been identified, on basis of which a two-stage collision model is proposed. Two magnetically very "soft" components are present in most of the samples examined. They represent (1) a recent field component, and (2) an eastwards directed component streaking towards the plane of the east-west-oriented foliation pattern.

Two other magnetically rather "soft" components are present in some of the samples only. One of these components (3) indicates a palaeolatitude of the sampled area at about 23° N and suggests crustal shortening in south-central Asia over more than 10° of latitude after final collision of the Ladakh island arc, then already united with the Indian subcontinent. The other soft component (4) may be related to younger phases of the Himalayan Orogeny (Mio-Pliocene or younger).

Two magnetically very "hard" components (5 and 6) indicate a palaeolatitude of the sampled area at 7–10° N. These components were acquired shortly before final closure of the isotopic systems, but after initial collision (pre-Middle Eocene) of the Indian subcontinent with the Ladakh island arc along the Indus-Tsangpo suture zone. This palaeolatitude of the sampled area is in good agreement with palaeolatitudinal control of the Indian plate from DSDP core data and indicates that after their initial collision, this combined Indian subcontinent–Ladakh island arc block moved northwards over a total distance of about 25° of latitude.

Declination control of the various magnetization components suggests that after initial collision the sampled area followed the counterclockwise rotation of the Indian plate, which subsequently reversed to a clockwise rotation after final collision with Asia.

1. Introduction

The impact of the northwards-drifting Indian subcontinent upon collision with southern Asia is heavily reflected in the structure and magnitude of the surrounding mountain chains (Fig. 1), most notably in the Hindu Kush-Pamir-Karakorum arc. The most evident relict of this collision is the Indus-Tsangpo suture zone [1–6]. This semi-continuous belt of thrust faults and ophiolitic rocks lies to the north of the Himalayas and swirls around the Western Himalayan Syntaxis via Kohistan into Baluchistan (Fig. 1, [2,7]).

The tectogenesis of this important zone has been interpreted variously by different workers (see review by Powell and Conaghan [8]). The various models differ mainly in estimates about the original northern extension of the Indian subcontinent ("Greater India") and the suggested presence of former Gondwanaland fragments to the north of the Indus-Tsangpo suture zone.

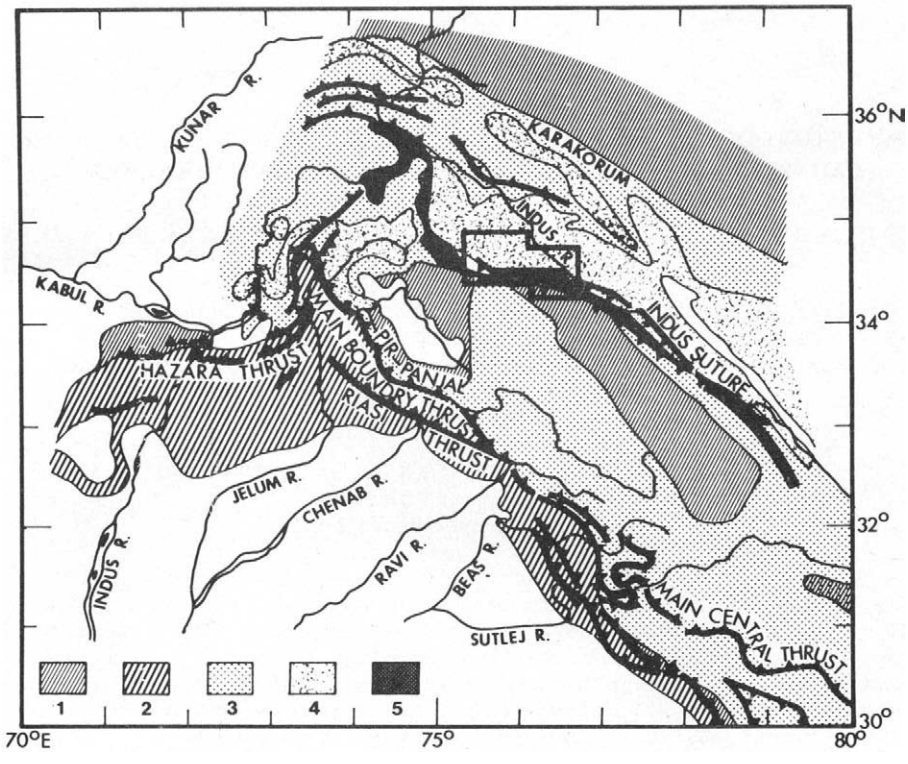


Fig. 1. Geological sketchmap of the northwest Himalayas (after Gansser [1] and Calkins et al. [84]. The sampling region is boxed in. Legend: 1 = middle and late Tertiary (post-collisional); 2 = Mesozoic and Tertiary up to Eocene (pre-collisional); 3 = Palaeozoic and Precambrian; 4 = granite; 5 = ophiolite.

(1) The more widely accepted class of models envisages the Indus-Tsangpo suture zone as the site of initial collision between southern Asia and the Indian subcontinent. Only moderate subduction of continental crust took place, and southwards progradation of "fracture" zones has occurred [5,9,10]. The Indus-Tsangpo suture zone became choked off after initial collision (pre-Middle Eocene) with a subsequently renewed convergence during the main Himalayan orogenic phase (Neogene) at the site of the Main Central Thrust (MCT) zone. From Late Miocene times onwards subduction may have shifted to the Main Boundary Fault (MBF) zone according to Le Fort [9]. Recent micro-earthquake activity studies in the Western Himalayan Syntaxis [11,12] do not only delineate the MCT which is very close to the MBT in this region, but may even indicate incipient thrust faulting parallel to but about 200 km further to the south of the MBF [11].

In this class of models the Tibetan Plateau is seen as an integral part of southern Asia, although island arc accretion may have occurred. The very evident indentation of India into Asia is variously interpreted as accommodated by ductile creep at deeper levels resulting in appreciable shortening at the surface of the Tibetan Plateau [13] or by lateral squeezing of parts of south-central Asia along east-west-aligned strike-slip faults, out of the way of the advancing Indian continental fragment [14–17]. A total crustal shortening in south-central Asia over more than 1000 km has been postulated by Molnar and Tapponnier [14] on the basis of ERTS photographic studies of the latter wrench faults.

(2) Other models favour a much larger original northern extent of the Indian subcontinent, and derive in part from stratigraphic [18,19] and marine geophysical evidence [20–23] for the former presence of a continental mass off western Australia which pre-

cludes the existence of the oceanic Sinus Australis gap in the original Smith and Hallam [24] Gondwanaland reassembly or subsequent revisions [25–27].

Crawford [28,29] postulated that the Tibetan Plateau forms an integral part of the Indian subcontinent, positioned before breakup of Gondwanaland against southwestern Australia, with the Tarim block possibly aligned along northwestern Australia. In his scheme the Indus suture zone represents a relic of a Permian to Jurrasic deep reaching rift zone which later closed.

Powell and Conaghan's [30] model explains the far-thicker than normal sialic crust beneath the Tibetan Plateau by postulating a formerly far-greater northern extent of the Indian subcontinent (i.e. the "Greater India" of Veevers et al. [18]), now underthrust under the Tibetan Plateau and possibly extending as far north as the Kun Lun–Astin Tagh–Nan Shan zone. In this scheme the Indus-Tsangpo zone is seen as the original suture between India and Asia which become inactive after collision of the two continental blocks (pre-Middle Eocene). They envisage large-scale continental underthrusting to have commenced further southwards along the MCT zone in post-Oligocene times. The MBF zone is seen in this model as part of a related "schuppen zone".

(3) Stöklin [31] proposed an interesting model, interrelating the main structural zones of Iran, Afghanistan and Baluchistan with the Pamir–Karakorum–western Himalayan region. Presence of former Gondwanaland fragments in central Iran and Afghanistan is well established now on geological [31–36] and palaeomagnetic grounds [37–39] as summarized by Stöcklin [31] and Klootwijk [40], and Stöcklin postulates existence of similar Gondwanaland fragments further to the east. In his view a series of continental fragments ("Central Domain"), now situated in a vast belt stretching from Iran in the west to the Tibetan Plateau in the east, broke off from the northern margin of Gondwanaland as early as latest Palaeozoic/early Mesozoic. These fragments drifted northwards to amalgamate with Eurasia prior to the Early Jurassic, thus leading to closure of the original northern "Palaeotethys". The southern "Neotethys", formed in the wake of these fragments, was then closed in the Late Cretaceous/early Tertiary when the main Gondwanaland fragments, i.e. India and Afro-Arabia, converged with these precursors.

The timing of the Palaeotethys movements is not too well dated as yet, and they may not necessarily have been synchronous. Initial fragmentation may have started in the Zagros area already during the Early Permian [33] but the main crustal separation probably occurred in the Late Triassic as Stöcklin [31] concludes from an important evolutionary break conspicuously present in the "Central Domain". Closure of the "Palaeotethys" had occurred in the west by the Early Jurassic but is not well defined in the east. Several suture zones have been identified in Tibet [41], but it is not clear whether they represent accretion of island arcs [6] or accretion of original Gondwanaland and fragments [31,41].

Clearly, the original southern border of Eurasia is not well defined as yet. For this, palaeomagnetic data from rocks to the north of the Indus-Tsangpo suture zone are badly needed. The only such data available to date are preliminary palaeomagnetic results from Upper Devonian pisolitic ironstones from Citral (northwest Pakistan), reported in part I of this series [42], and from Upper Cretaceous to Palaeogene red beds from south-central Tibet [43,44]. There is no field access to Tibet as yet, but the Indus-Tsangpo suture zone and its westwards continuation can be studied by foreigners in various parts of northwestern India and northern Pakistan. During a short sampling trip in October 1976 to the Indus-Tsangpo suture zone in Ladakh, we sampled the Ladakh Intrusives at Kargil and the Indus Molasse and the Dras Flyschoids at Khalsi-Lamayura. Results from the Ladakh Intrusives are presented here.

2. Geological setting

The overall geological structure of central Ladakh (Figs. 1, 2) has been described recently in a number of studies [6,45–49]. General consensus exists about the main structural trends, but there is considerable and often confusing disagreement about the stratigraphic classification and the age ranges of the various formations. In particular the rather literal and on occasions inappropriate use of the genetically related terms Molasse and Flysch has resulted in a confusing variety of formation names.

In central Ladakh a number of zones generally parallel to the ENE-WSW-trending Indus-Tsangpo

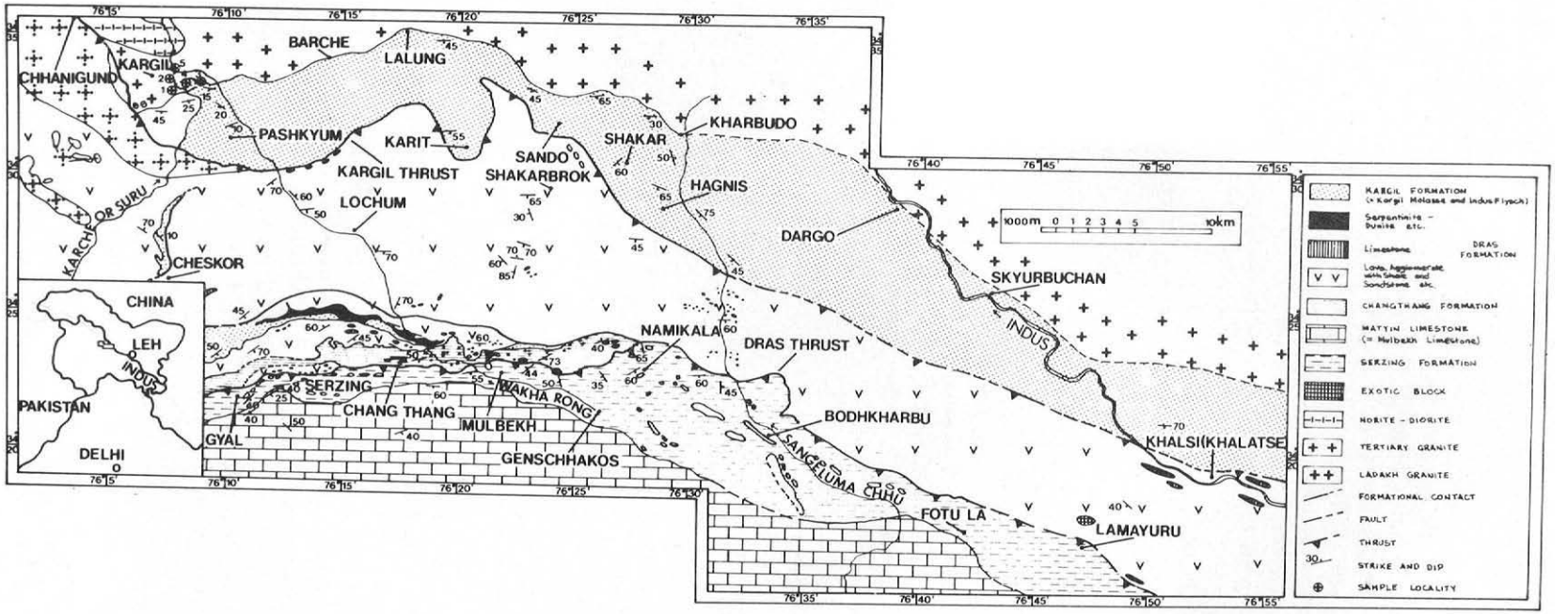


Fig. 2. Geological sketchmap of the sampling region (after M.L.S.).

suture zone can be distinguished (Fig. 2) and they will be described from north to south.

2.1. The Ladakh Intrusives ("Ladakh Granite")

Immediately to the north of the Indus-Tsangpo suture zone stretches the vast belt of the Ladakh Intrusives ranging in petrographic character from tonalites to granodiorites and granites. Frank et al. [6] believe that the whole intrusive suite belongs to one major plutonic event of Late Cretaceous to Eocene age. In constrast, Shah et al. [48] distinguish between what they believe to be an older pre-Tertiary granite ("Ladakh Granite") and a younger "Tertiary Granite". Along its southern margin the older suite of intrusives are stratigraphically overlain by conglomeratic-arkosic sediments (Lower Indus Formation of Ravi Shankar et al. [47]; Indus Molasse of Frank et al. [6]) for which a lower age as old as Albian has been inferred [47].

Radiometric age determinations on the Ladakh Intrusives group around 49–45 m.y. (Rb-Sr method: 45 m.y. by Desio and Zanettin [50]; K-Ar method: 48.4 ± 1.7 m.y. on biotite by Saxena and Miller [51] and a range of ages between 45 and 48 m.y. according to Geological Survey of India [52] and le Fort [53]; fission track dating: 46 ± 5 m.y. on apatite, and a range of ages between 35 and 15 m.y. by Lal and Nagpaul [54]). Whole rock Rb-Sr determinations on samples from our collection were not feasible because of an insufficient spread in $^{87}Rb/^{86}Sr$ ratio (0.103–0.53).

The grouping of the ages obtained from the various methods indicates a rather fast closing of the various isotopic systems. These ages probably are indicative only for cooling of the batholith after initial collision of the Indian continental fragment with a northern continental mass. A much older age of emplacement of the intrusives can be inferred from the age of the overlying sediments.

It is most likely that these Ladakh Intrusives represent a former island arc off south-central Asia [6,55–57] formed by accumulation of intrusives as a result of consumption of former oceanic crust since the Early Cretaceous separation of the Indian continental mass from Gondwanaland [19,20,22]. Geochemical work by the two of us (S.K.S. and J.G.) on the Dras Volcanics indeed indicates the presence of an island arc volcanic suite evolving from a tholeiitic composition at the base to a shoshonitic composition at the top, suggestive for a steeply dipping subduction zone [58,59].

2.2. The Indus Molasse

(Kargil Formation of Shah et al. [48], Indus Formation of Ravi Shankar et al. [47]). Facies variations and strong deformation occur in this zone leading to confusion of this molasse with flysch as summarized by Shah et al. [48]. Frank et al. [6] distinguish between a northern autochthonous belt of post-Early Cretaceous age and transgressive on the Ladakh Intrusives, and a southern parautochthonous belt (Hémis conglomerates) which is in part of Eocene age. Ravi Shankar et al. [47] date the stratigraphically lower part of this northern zone (Lower Indus Formation in their terminology) as Albian to Maastrichtian with a Palaeocene to Middle Eocene age for the Upper Indus Formation.

The southern limit of this zone is bound by a major north-directed thrust, the Kargil thrust of Shah et al. [48], along which a major tectonic unit comes into juxtaposition with the Indus Molasse. Further southwards this tectonic unit is in turn overthrust to the north (Dras thrust of Shah et al. [48], Zaskar thrust of Frank et al. [6]) by Palaeozoic and Mesozoic rocks of the Zaskar Ranges. This important tectonic unit is of a very complex nature. Two major lithologic units can be discerned, i.e. the Dras Volcanics in the north and the Indus Flysch (Zojila Formation of Shah et al. [48]) to the south.

2.3. The Dras Volcanics

These form one of the major geological features of the Ladakh area. The volcanic facies of the Dras Volcanics, which is so predominant in the western Ladakh area is gradually overtaken by a sedimentary facies further to the east. Ophiolitic melange belts border the Dras Volcanics to the north and south. These belts are directly related to steep thrust faults, with recently discerned glaucophane schists [6]. This entity of ophiolitic melange and Dras Volcanics (i.e. the Samdo Formation of Ravi Shankar et al. [47] and the Dras Formation of Shah et al. [48] is of a Middle to Late Cretaceous age [47,48], possibly ranging into the Early Eocene [48].

2.4. The Indus Flysch

Two types were distinguished by Frank et al. [6] in the Indus Flysch zone, differentiated lithologically and by age, i.e. the Namika La Flysch and the Lamayura Flysch. No clear separation of these types could be made by them in the field. The Lamayura Flysch which has been recognized so far only near its type locality north of the Lamayura monastery, may range from Triassic to Early Jurassic in age. The age of the more prevalent Namika La Flysch (Zojila Formation of Shah et al. [48]) is disputed. Frank et al. [6] assume a Cretaceous to Palaeogene age. However, recently one of the authors (M.L.S.) has found *Daonella* cf. *Indica* (Ladinian to Carnian) from the Namika La Flysch near Namika La, which is also characteristic for the Lamayura Flysch [6]. So both flysch types may represent time equivalent units of different facies. This finding supports Shah et al.'s [48] suggestion for a Triassic to possibly Permian age for the whole Zojila Formation, based on an inferred autochthonous nature of conspicuous "exotic" limestone blocks of Triassic [6,48,60] to Permian age [48]. Sharma and Shah [61] also reported an Upper Permian fauna from exotic blocks within the Dras Volcanics north of Lamayura, but there is no lithological nor faunal relationship with the "exotics" of the Namika La Flysch.

In the sampled area near Kargil the overall ENE-WSW-directed structural trend of central Ladakh is interrupted by a southward spur of the Ladakh Intrusives (Fig. 2). Here, the structure is further complicated by the presence of a large noritic-dioritic body, by felsic and mafic dykes, and by northwards directed thrusting (Kargil thrust) within the Ladakh Intrusives, with metamorphosed Dras Volcanics forming a marked mylonite zone within the intrusives. According to Shah et al. [48] the overriding "Tertiary Granite" and the underlying main body of the "Ladakh Granite" which has been sampled in this study, represent two separate episodes of igneous activity. The two granite bodies are of different petrological character, the overriding granite being a hornblende granite, darker in colour than the underlying biotite-hornblende granite and granodiorite of the "Ladakh Granite". The main difference between the two granites lies in the relative grade of metamorphism of the "Ladakh Granite" which has been rendered gneissose at many places. In contrast, the overlying "Tertiary Granite" is fresh without any evidence of metamorphism. However, radiometric age determinations have not substantiated as yet such an inferred distinctly older age of the "Ladakh Granite". According to Frank et al. [6] the regional metamorphism is of subgreenschist facies only, so obtained absolute ages which range between 49 and 45 m.y. may represent cooling ages of the sampled units subsequent to emplacement, rather than overprinted ages, caused by a regional reheating upon initial collision of the Indian subcontinent. Frank et al. [6] argue that the whole intrusive suite of the Ladakh Intrusives does belong to one major plutonic event.

An upper age limit for the "Ladakh Granite" at Kargil is given by the age of the discordantly overlying Indus Molasse, which has been referred to in the sampling region as Kargil Formation [48] and Ladakh Molasse [62]. Carbonate pebbles in the uppermost part of the well exposed molasse sequence at Pashkyum (near Kargil), are of Early Eocene to early Middle Eocene age [6]. Although Tewari [62] considers part of the molasse to be of Mio-Pliocene age, there seems to be consensus for an Eocene age of at least part of the Indus Molasse sequence at Kargil [46,48,63]. In the Kargil area the Indus Molasse shows a rather constant dip of about 20–25° south, with minor warping related to the Kargil thrust.

3. Sampling and method of analysis

In the area near Kargil (Fig. 2, Table 1, 76°7′E, 34°34′N) we collected 41 block samples in granodioritic rocks from four localities in the "Ladakh Granite" (sites 1, 2, 3 and 5). In addition 13 block-samples were taken from a 75-m-thick doleritic dyke (site 4) intrusive into the "Ladakh Granite". Thin porphyritic dykes intrude the granodiorite at sites 1 and 2. At site 5 mafic xenoliths have been assimilated partly into the granodiorite. All rocks, apart from site 3 are of a very fresh appearance. At the latter site we collected only two samples from a position a few meters underneath the lower Tertiray erosion disconformity with the overlying Indus Molasse.

Orientation of the samples was done with an orientation device described by Embleton and Edwards [64]. Unfavourable weather conditions prevented use of a solar compass, so all readings were done with a magnetic compass, cross checked with distant bearings.

TABLE 1

A. Initial natural remanent magnetization data, Ladakh Intrusives at Kargil (76°7'E 34°34'N)

Locality	N (specimens)	Intensity (mA m^{-1})	Induced (mA m^{-1}) *	Applied dip correction	Rock type
1. IKGD 1-12	11 (44)	$2 \times 10^3 - 2 \times 10^4$	$4 \times 10^5 - 8 \times 10^5$	N80°E/35°S	granodiorite
2. IKGD13-27	15 (60)	$2 \times 10^2 - 3 \times 10^3$	$2 \times 10^5 - 3 \times 10^5$	N90°E/20°S	granodiorite
3. IKGD28-29	2 (8)	$3 \times 10^3 - 9 \times 10^3$	$4 \times 10^5 - 6 \times 10^5$	N100°E/15°S	granodiorite
4. IKGD30-43	14 (56)	$2 \times 10^4 - 6 \times 10^4$	$4 \times 10^5 - 8 \times 10^5$	N70°E/25°S	doleritic dyke
5. IKGD44-56	13 (52)	$4 \times 10^2 - 5 \times 10^3$	$1 \times 10^5 - 6 \times 10^5$	N90°E/20°S	granodiorite

B. After alternating field and/or thermal demagnetization

Locality	Component	N	Direction before dip correction				Direction after dip correction	
			D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)
1	6	11	183.5	+9.0	38.2	7.5	183.9	−15.3
3	6	2	358.8	+5.4	50.7	–	358.1	+20.1
4	6	14	2.7	−9.9	72.1	4.7	3.0	+13.2
1	5	9	169.7	+5.5	10.1	17.1	169.6	−19.6
3	5	2	353.7	+16.6	2.9	–	351.7	+31.0
4	5	12	1.4	−4.6	37.5	7.2	0.2	+18.2
2	4	13	7.9	+20.8	15.7	10.8	9.7	+39.7
5	4	13	5.2	+21.9	10.8	13.2	6.5	+41.8
2	3	13	16.4	+32.1	12.5	12.2	23.3	+50.5
5	3	13	5.9	+27.3	12.6	12.1	7.7	+47.1

C. Mean components results

Component	N	D (°)	I (°)	k	α_{95} (°)	South pole position	dp	dm	Palaeolatitude
6	27	3.0	+14.6	53.4	3.8	62.7°S, 69.7°E	2.0°	3.9°	7.4°N
5	23	354.9	+19.4	13.6	8.5	64.9°S, 88.0°E	4.7°	8.9°	10.0°N
4	26	8.2	+40.8	13.3	8.1	76.7°S, 41.8°E	5.9°	9.8°	23.3°N
3	26	17.1	+49.1	12.4	8.4	74.9°S, 0.7°W	7.3°	11.1°	30.0°N

* Induced magnetization in 0.1 mT applied field.

In the laboratory, 467 specimens (2.5 cm diameter and 2.25 cm in height) were cored out of the samples. All natural remanent magnetization (NRM) measurements were done on a Digico fluxgate spinner magnetometer and susceptibility measurements were done on a Digico bulk susceptibility unit. On the basis of the initial NRM measurements one specimen per sample was selected for progressive thermal demagnetization in 17 steps up to 580°C in a furnace described by McElhinny et al. [65]. Another specimen from each sample was progressively demagnetized in 12–24 steps in alternating fields up to 200 mT using a two-axis tumbler modified after Evans [66]. Another two specimens per sample were subjected to partial progressive alternating field (AF) demagnetization in 8–13 steps over a selected range of AF intensities. Half of the specimens were demagnetized using the two-axis tumbler, the remainder of the specimens were demagnetized "statically" along three orthogonal axes [67,69]. Analysis of specimen directions was done according to Zijderveld's method [69,70] using orthogonal projection figures.

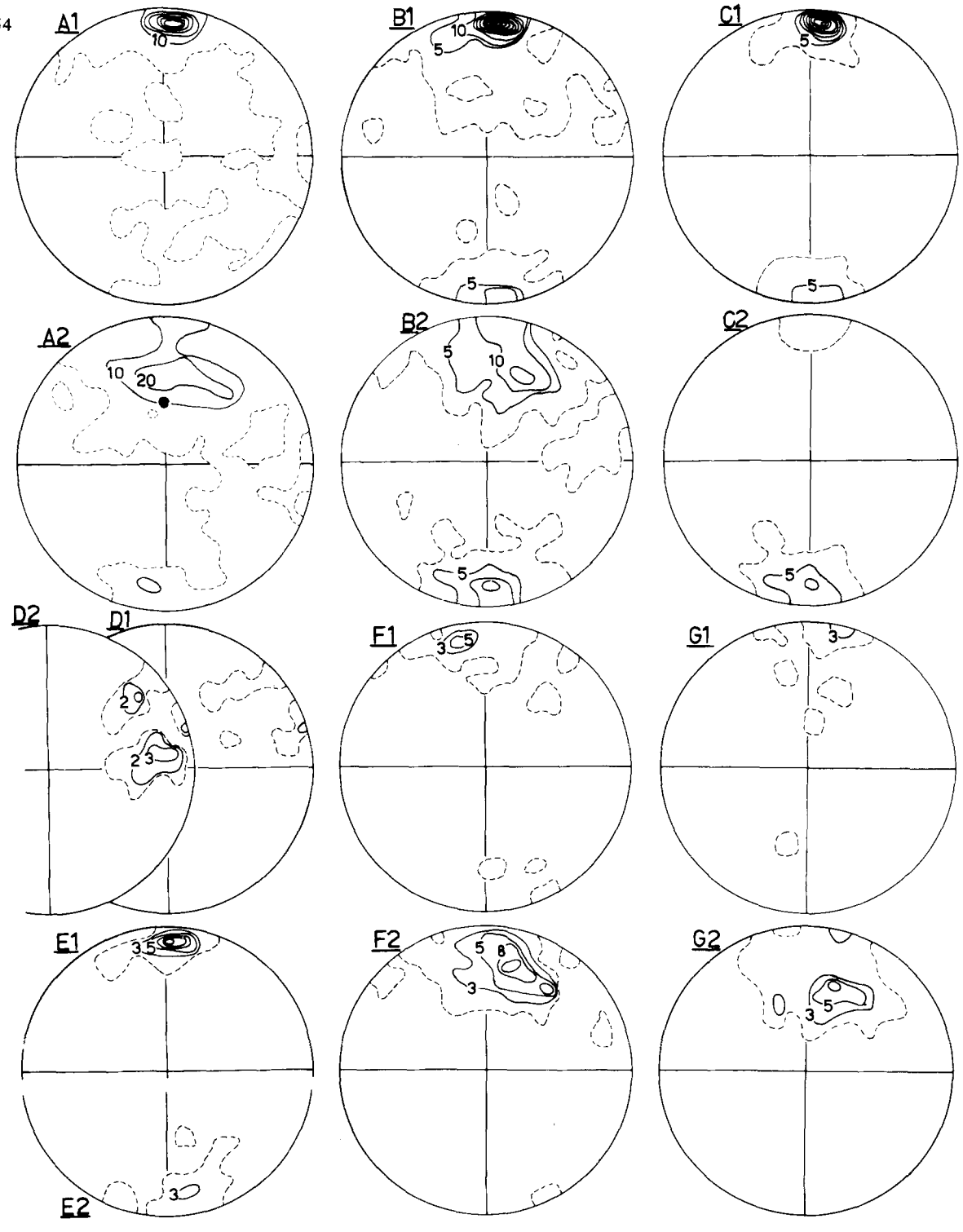


Fig. 3. Frequency distribution of magnetic directions of individual specimens in equal area projection, counted with a 1% circle. Subscript 1 denotes upper hemisphere projections, subscript 2 denotes lower hemisphere projections. Frequency intervals as indicated in the individual plots. A. Initial directions of the natural remanent magnetization. The full dot indicates the present field direction at the sampling area ($D = 2°, I = +52°$). B. Directions from components 2, 3, 4, 5 and 6 combined for all specimens. C. Component 6. D. Component 2. E. Component 5. F. Component 3. G. Component 4.

4. Results

Initial intensities of NRM ranged from 4×10^2 mA m^{-1} to 6×10^4 mA m^{-1}, whereas initial intensities of induced magnetization (0.1 mT applied field) were generally 1 to 2 orders of magnitude higher (1×10^5 mA m^{-1} to 8×10^5 mA m^{-1}).

The initial specimen directions showed a dual polarity of directions. The directions of specimens from site 2, 3, 4 and 5 showed normal polarity directions, northwards pointing with low upwards to moderate downwards inclination (Fig. 3A). This streaking towards downwards inclinations is clearly due to a bias towards the present field direction at the sampling area ($D = 2°, I = +52°$). The directions of specimens from site 1 were mainly of reversed polarity, southwards pointing with low upwards to moderate downwards inclination (Fig. 3A). Both groupings also revealed a streaking towards easterly directions.

Thermal demagnetization of the pilot specimens showed two types of magnetic behaviour:

(A) Specimens from sites 1, 3 and 4 showed only a minor reduction in intensity up to 500°C, but displayed a very discrete blocking temperature range between 550° and 580°C (Fig. 4A).

(B) In contrast, specimens from sites 2 and 5 showed a gradual intensity decay up to 500–550°C, at which temperatures generally less than 20% of the initial intensity remained, followed by a fast decrease between 550° and 580°C (Fig. 4B).

Upon AF demagnetization, group A samples showed magnetically "harder" characteristics with a broad coercivity spectrum ranging up to 200 mT (Fig. 4C). Group B samples showed only magnetically "softer" characteristics with coercivity values not higher than 30–50 mT (Fig. 4D).

Throughout the sample collection, six different magnetization components have been discerned, which could be identified equally well upon AF and thermal demagnetization (Figs. 4, 5):

(1) In the samples from both group A and group B a northwards-and downwards-pointing component, evidently of recent origin, was removed generally in alternating fields of 10–15 mT or below 300–400°C upon thermal treatment (Figs. 5F, G, K, L, M, T, U).

(2) Another northeast-to eastwards-pointing component of low to moderate downwards inclination (Fig. 3D) was broken down in the same alternating field or thermal interval (Figs. 3D, 5B, C, P, Q, R, S, V), but resided occasionally also in magnetically harder domains (Fig. 5S). Composite breakdown of this eastwards-directed component (2) and the recent field component (1) was observed in various specimens.

Another two magnetization components could be identified in the magnetically "softer" specimens of group B.

(3) A north-northeastwards- and downwards-pointing magnetization component (Figs. 3G, 5D, E, L, M, R, S, X) could be removed upon thermal treatment between 100°C and up to 500°C, but was generally less well separable upon AF demagnetization (site 2: 0–25 mT; site 5: 0–10 mT).

(4) Another "soft" magnetization component, northeastwards pointing and of shallow inclination (Figs. 3F, 5D, E, F, U) was broken down generally above 400°C and most evidently between 500° and 580°C. Upon AF treatment this component was broken down at the higher field intensities (site 2: 20–60 mT; site 5: 10–30 mT).

Finally, another two magnetization components

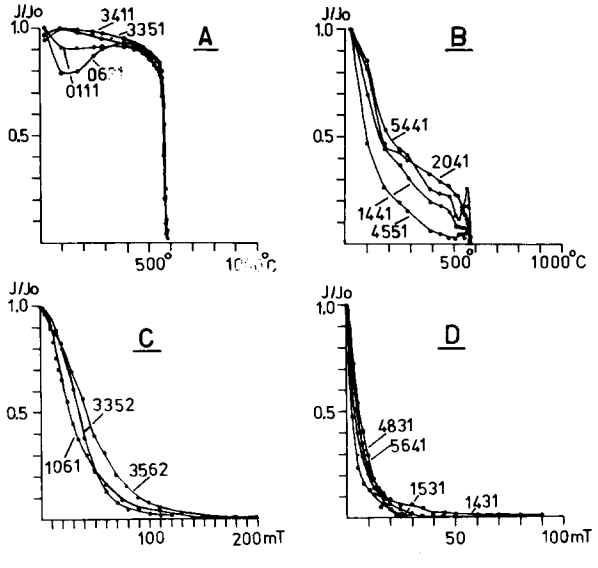
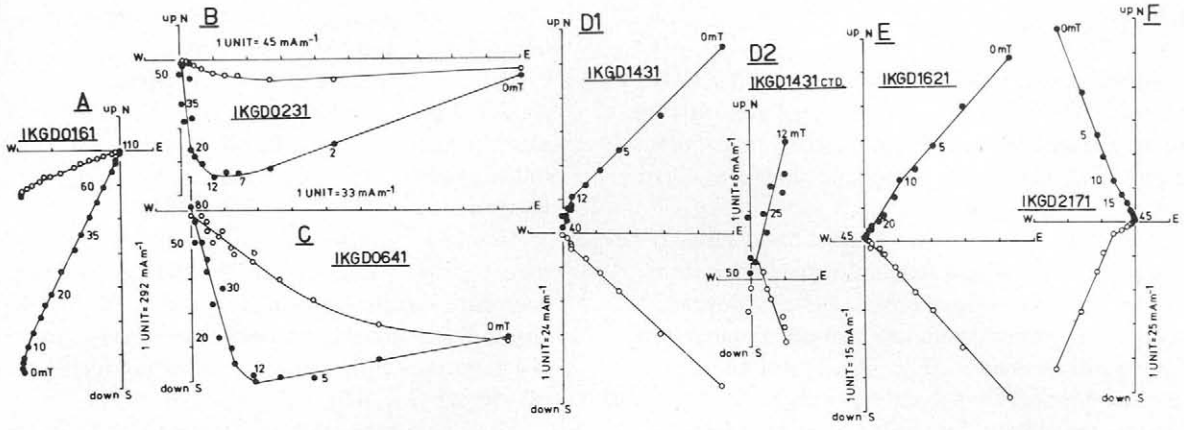
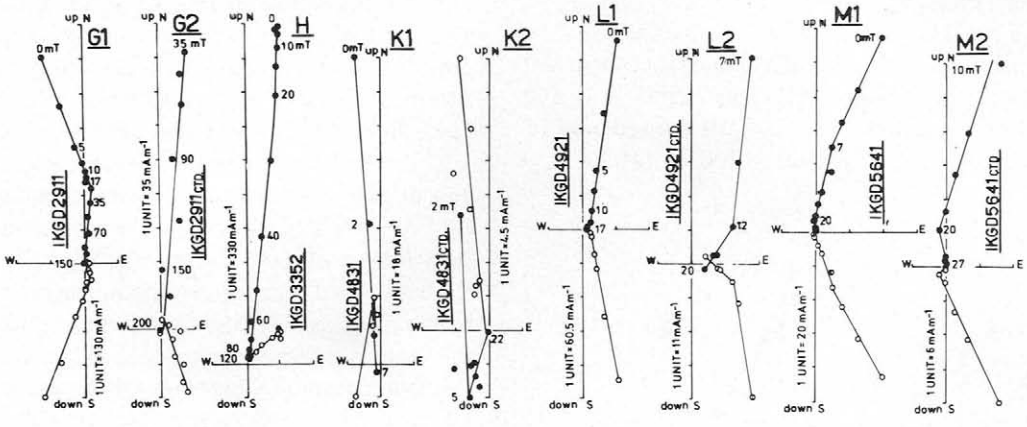
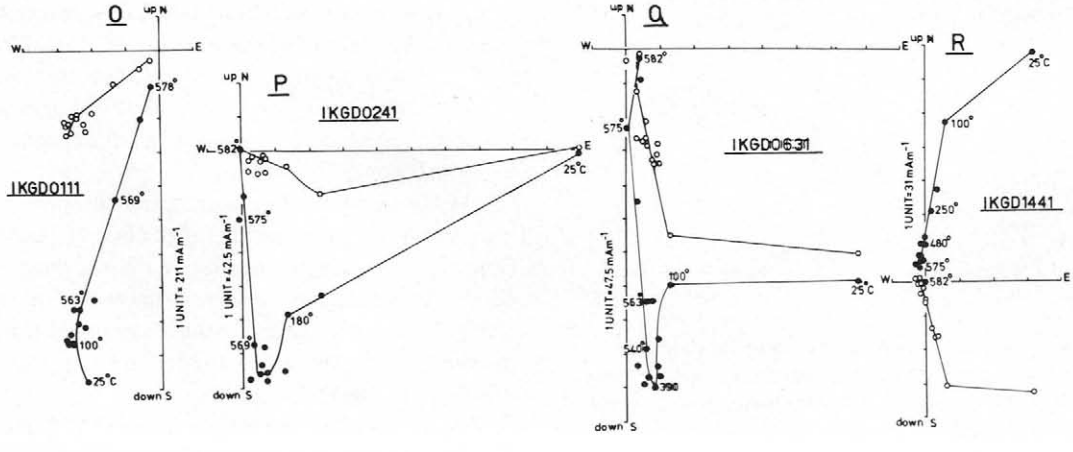


Fig. 4. Normalized intensity decay curves of the remanent magnetization during thermal treatment (A: group A samples, B: group B samples) or during AF demagnetization (C: group A samples, D: group B samples).


A
IKGD0161
B
IKGD0231
1 UNIT= 45 mAm⁻¹
1 UNIT= 33 mAm⁻¹
C
IKGD0641
1 UNIT= 292 mAm⁻¹
D1
IKGD1431
D2
IKGD1431 CTD.
1 UNIT= 24 mAm⁻¹
1 UNIT= 6 mAm⁻¹
E
IKGD1621
1 UNIT= 15 mAm⁻¹
F
IKGD2171
1 UNIT= 25 mAm⁻¹
up N
down S
W.
E
0mT


G1
IKGD2911
1 UNIT= 130 mAm⁻¹
G2
IKGD2911 CTD.
1 UNIT= 35 mAm⁻¹
H
IKGD3352
1 UNIT= 330 mAm⁻¹
K1
IKGD4831
1 UNIT= 18 mAm⁻¹
K2
IKGD4831 CTD.
1 UNIT= 4.5 mAm⁻¹
L1
IKGD4921
1 UNIT= 60.5 mAm⁻¹
L2
IKGD4921 CTD.
1 UNIT= 11 mAm⁻¹
M1
IKGD5641
1 UNIT= 20 mAm⁻¹
M2
IKGD5641 CTD.
1 UNIT= 6 mAm⁻¹
up N
down S
W.
E


O
IKGD0111
1 UNIT= 211 mAm⁻¹
P
IKGD0241
1 UNIT= 42.5 mAm⁻¹
Q
IKGD0631
1 UNIT= 42.5 mAm⁻¹
R
IKGD1441
1 UNIT= 31 mAm⁻¹
up N
down S
W.
E

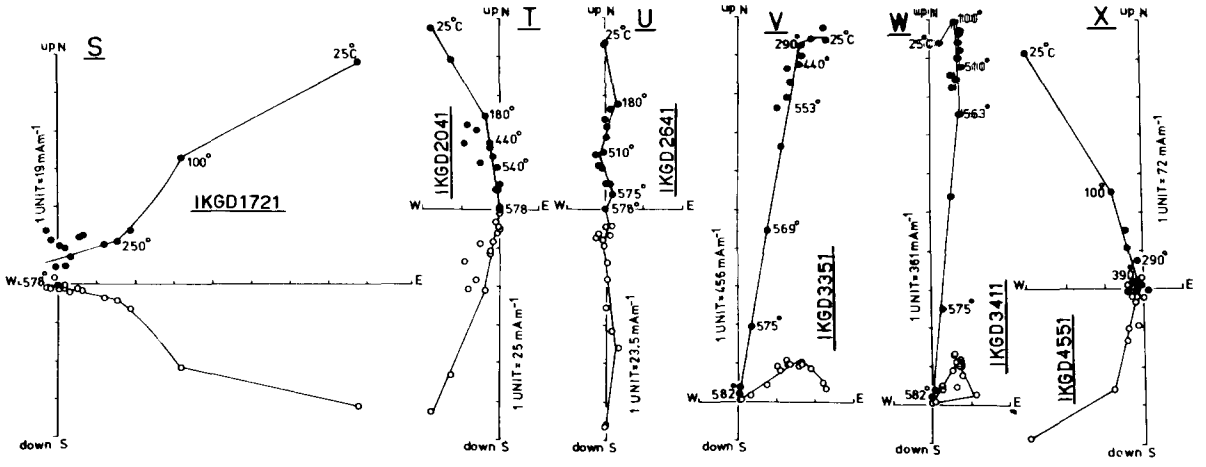


Fig. 5. Demagnetization diagrams of specimens treated by AF field demagnetization (A–N) or by thermal methods (O–X). The points denote successive positions (in orthogonal projection) of the end of the resultant magnetization vector during progressive demagnetization. Open circles denote projections on the vertical east-west plane. Full circles denote projections on the horizontal plane. Numbers denote successive peak values of the applied alternating fields or the applied temperatures.

were distinguished in the magnetically "harder" specimens of group A. Site 1 specimens showed reversed polarities for both components, whereas sites 3 and 4 showed normal polarities.

(5) A slightly north-northwest- (site 3 and 4) to south-southeast- (site 1) directed component of shallow inclination (Figs. 3E, 5G, H, P) was thermally broken down in site 1 and 3 specimens, between 300° and 500°C and occasionally up to temperatures as high as 580°C, but was not evidently present in site 4 specimens. Upon AF demagnetization this component was broken down in site 3 and 4 specimens, between 10 and 60 mT but was not well identifiable in site 1 specimens.

(6) Another "harder" magnetization component, north- to north-northeast- to south-southwest-directed and of shallow inclination (Figs. 3C, 5A, B, C, D, Q, V, W) became thermally broken down between 550° and 580°C. Upon AF demagnetization his component was the main one to be observed and could be broken down between 20 and 200 mT. However, in presence of component 5, this component 6 generally could be broken down between 40 and 100 mT.

5. Interpretation

The main magnetization carrier in group A samples is a very pure magnetite as indicated by the very discrete blocking temperature spectrum, ranging from 550° to 580°C. The distributed blocking temperature spectrum of group B samples, in contrast, indicates a titanomagnetite as the main magnetization carrier, but a rather pure magnetite phase seems also to be present. Polished thin sections show the presence of a fresh magnetite in samples from both group A and group B with only a slight martitization in samples from the latter group. The presence of secondary ironhydroxides was observed only in the two samples from site 3, which is obviously related to their position just underneath the lower Tertiary erosion surface.

Knowledge of the time acquisition of the various magnetization components is crucial to their interpretation. Parts of the Ladakh Intrusives were emplaced probably already in pre-Albian times as can be inferred from the normal sedimentary contact between the "Ladakh Granite" and the overlying (Northern) Indus Molasse. Reworked fossils in the basal conglomerate directly overlying the eroded "Ladakh Granite" at Upshi in the eastern Ladakh region range from Permian to Early Cretaceous in age with conspicuous absence of post-Mesozoic fossils [6]. Ravi Shankar et al. [47] data the basal part of this sedimentary unit (Lower Indus Formation in their terminology) as Albian to Cenomanian in age on basis of its faunal assemblage. In the sampling area near Kargil, however, fossil content of the molasse directly overlying the "Ladakh

Granite" has not indicated so far ages older than Eocene [46,63], whereas carbonate pebbles in the top part of the molasse sequence at Pashkyum suggests a lower Eocene to lower Middle Eocene age [6]. Radiometric datings on the Ladakh Intrusives in this area, obtained by the Rb-Sr and the K-Ar method concentrate between 49 and 45 m.y. and indicate a period a rather fast cooling and closure of the isotopic systems. Fission track datings in the Kargil area range between 35 and 15 m.y., but a single apatite age of 46 ± 5 m.y. [54] reported from a locality very near to the sampling are may suggest that the area has not been affected by any major thermal event since then.

Frank et al. [6] observed a regional metamorphism in central Ladakh not higher than of subgreenschist facies. Petrographic analysis of thin sections of our samples from the Kargil area and SEM studies indicate presence of an intermediate plagioclase An% 30–63, newly formed epidote, biotite and actinolithic hornblende, and the partial or total replacement of pyroxene by cummingtonite. Although these observations in themselves may indicate a metamorphic event as high as the epidote-amphibolite facies [71], the observed low-grade regional metamorphism [6] may suggest that these reactions occurred during extended cooling of the igneous complex [72]. Thus it is possible that the period of rather fast cooling (49 ot 45 m.y.) as indicated by the radiometric datings, only shortly post dates initial emplacement of the "Ladakh Granite" in the Kargil area and did not follow a period of reheating upon continental collision of an already earlier emplaced intrusive.

The above arguments are of academic interest only, with respect to the time of acquisition of the "harder" magnetization components 5 and 6. An epidote-amphibolite facies metamorphism, if it occurred at all locally in the sampled area, must have involved temperatures in excess of 500°C, and would have resulted in a total resetting of even the hardest magnetizations observed [73]. Hence, the radiometric age determinations can be taken as indicative of the age of at least the harder magnetization components.

Assuming that the 20–25° southwards dip of the Molasse beds at Kargil accounts for total accumulated dip of the "Ladakh Granite" after cooling of the intrusives, a similar dip correction was applied to components 3, 4, 5 and 6. The obtained mean directions are given in Table 1 and the corresponding pole positions and palaeolatitudes for the Kargil area are plotted in Figs. 6 and 7. Clearly these palaeolatitudes are all to the south of the present latitude of the sampling area, with components 5 and 6 indicating a farther southern palaeoposition than components 3 and 4. A continuing northward movement during late Mesozoic and Cenozoic times of the Indian plate with subsequent indentation of south-central Asia is well established nowadays. Regardless of which plate (or subplate) the

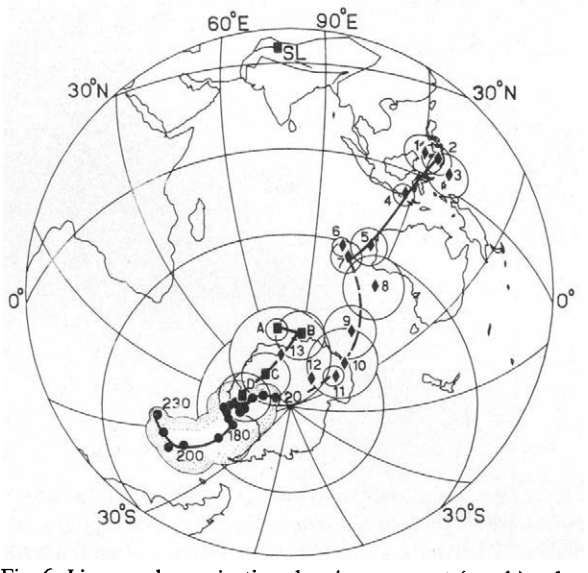


Fig. 6. Linear polar projection showing apparent (south) polar wander paths for India (diamonds after Klootwijk [40]) and Eurasia (dots, at 10-m.y. intervals, after Irving [74]), combined with the here obtained pole positions for the Ladakh Intrusives. Legend: 1 = Panchet beds (Upper Permian/Lower Triassic); 2 = Mangli beds (Upper Permian/Lower Triassic); 3 = Kamthi beds, Wardha Valley (Upper Permian/Lower Triassic); 4 = Rajmahal Traps (100–105 m.y.); 5 = 65–75 m.y., mean pole position from DSDP core results from the Indian plate; 6 = Deccan Traps (60–65 m.y.), lower reversed epoch; 7 = Deccan Traps (60–65 m.y.,), upper normal epoch; 8 = 55–65 m.y. mean pole position from DSDP core results from the Indian plate; 9 = 45–55 m.y., as above; 10 = 35–45 m.y., as above; 11 = 25–35 m.y., as above; 12 = 5–22.5 m.y., as above; 13 = Murree-Siwalik sediments, Potwar Basin (Middle-Upper Miocene). *A* = pole position for the Ladakh Intrusives according to magnetization component 6; *B* = as above, according to magnetization component 5; *C* = as above, according to magnetization component 4; *D* = as above, according to magnetization component 3. Ellipses of confidence (95% level) are shown for pole positions from the Indian plate and from the Ladakh Intrusives. For the Eurasian APW only the envelope of the ellipses is shown.

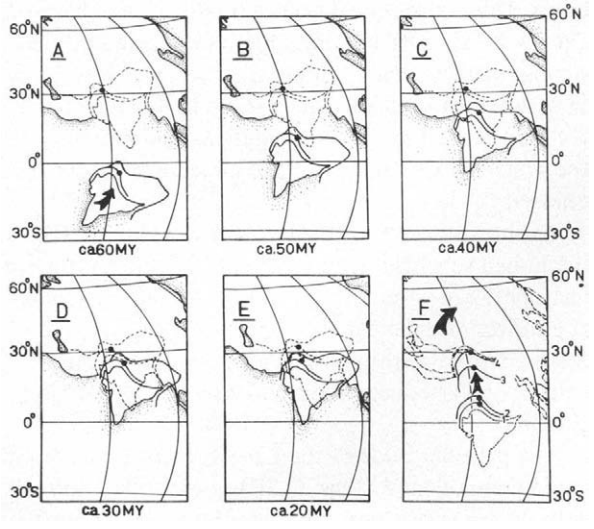


Fig. 7. A–E show successive positions, at 10-m.y. intervals, for the Indian subcontinent, as based on DSDP core data [40,80,82], and for southern Eurasia [74]. The former southern boundary of Asia is not well defined. For reference purpose only, the present position of India has been dashed in the Asian outline. The three zones north of India represent respectively: Baluchistan Himalayan–Arakan Yoma front, the Indus-Tsangpo suture zone and the northern boundary of "Greater India" according to Veevers et al. [18]. The dots represent the sampled locality of the here studied Ladakh Intrusives in their present position relative to both respectively the Indian subcontinent and to Asia. Fig. 7F shows a comparison of successive palaeolatitudinal positions of the sampling area (full dots), as determined from magnetization components 3, 4, 5 and 6 and indicated here respectively by positions *4*, *3*, *2*, *1*, with southern Eurasia positions, about 50 m.y. ago (dashed line) and about 20 m.y. ago (dash-dot line). To enhance visibility, the present outline of the Indus-Tsangpo suture zone (full line) and the outline of the Indian subcontinent (dashed line, only according to magnetization component 6) are shown, in their present form and position relative to the sampling area.

Ladakh Intrusives belonged to, the magnetically "harder" characteristics of components 5 and 6 and the observed southerly polaeolatitudes identify these components as the oldest ones observed in the present sample collection. Acquisition of these magnetic components shortly predated the closure of the Rb-Sr and the K-Ar systems between 49 and 45 m.y. The "softer" magnetization components 3 and 4 are clearly of a later origin, and are possibly related to younger phases of the Himalayan Orogeny.

The origin of the magnetically soft component 2 is not totally clear. It may well be that this eastwards-directed magnetization represents the effect of recent low-temperature oxidation with the newly formed magnetic phase preferentially aligned towards the foliation pattern.

6. Discussion

Comparison of the present palaeomagnetic results from Ladakh with palaeomagnetic data from the Indian plate and from Eurasia may put additional constraints on the various collision models and on the concept of "Greater India". Palaeomagnetic data from northern Eurasia, compiled recently by Irving [74], indicate a rather stable position during the late Mesozoic and early Cenozoic with only minor northward movement and a slight clockwise rotation during the last 50 m.y. (Figs. 6, 7). Palaeomagnetic data from the Indian subcontinent are very scanty for this period, and reliable results are presently confined to data from the Rajmahal Traps (100–105 m.y.) and the Deccan Traps (60–65 m.y.) as reviewed by Klootwijk [40]. This is to be regretted as the Indian plate underwent a fast northward movement of large magnitude during the latest Mesozoic and the earliest Cenozoic. Detailed knowledge about the palaeolatitudinal position of the Indian plate during this period is essential for evaluation of the various collision models.

The movement of the Indian plate relative to Australia is rather well detailed nowadays from marine magnetic anomaly studies [55,75–78], but for geographical positioning of the Indian plate such studies relied on the Australian apparent polar wander (APW) curve [79]. This indirect procedure seems to lead to a position of the Indian plate erroneously too far north, as has been pointed out recently by Peirce [80] and Klootwijk and coworkers [40,81], most probably due to inaccuracies in the Australian APW. Clearly more late Mesozoic and Cenozoic palaeomagnetic data from the Indian subcontinent itself are needed for absolute positioning of the Indian plate. In the absence of such data the best procedure is to use palaeomagnetic results from DSDP cores from sites on the Ninety East Ridge and from the Arabian Sea [80,82]. Unfortunately, these data suffer from absence of declination control, and straightforward computation of the arithmetic mean of individual

inclination values would consistently underestimate the true palaeoinclination. Peirce [82] designed an algorhythm to correct for this effect and a good estimate of the palaeolatitude of the sample site can be obtained in this way. The missing declination values can be simulated from the sequence of eulerian rotation poles (cf. [55]) obtained from analyses of the marine magnetic anomaly patterns. We have combined the simulated pole positions obtained in this way for the Indian plate (Fig. 6, [40]) for the age ranges: 65–55 m.y., 55–45 m.y., 45–35 m.y., 35–25 m.y., and 22.5–5 m.y. The resulting palaeolatitudinal positions and orientations of India are shown in Fig. 7.

The timing of collision of the Indian subcontinent with southern Asia is diputed. On the basis of sea-floor spreading data and geological evidence Powell and Conaghan ([18,30], see also Powell [55]) have claimed that initial collision occurred in pre-Middle Eocene times (about 53 m.y. ago). This was followed by a marked drop in spreading rates between India and Australia [8,55,76,77], and a reduction in the rate of northward movement of the Indian subcontinent from an initial 15–20 cm/yr to no more than 4–6 cm/yr [55]. In contrast, Molnar and Tapponnier [14] together with Dewey and Burke [13] consider that intimate contact between the Indian subcontinent and southern Asia did not occur until about the Eocene/Oligocene time boundary (about 38 m.y. ago) when the rate of relative movement slowed down, according to them, from about 10–18 cm/yr to about 5 cm/yr. This estimate, however, is up to refinement as Molnar and Tapponnier used average convergence rates over long periods of time.

The DSDP core data show that at about 50 m.y. ago the Indus-Tsangpo suture zone on the northern boundary of the Indian plate was still far off southern Asia (Fig. 7), even if following Powell's [55] assumption that its southern boundary was represented by the eastwards continuation of the Zagros-Makran zone. It is a significant observation, however, that the contemporaneous palaeolatitude of the Ladakh Intrusives, as obtained in the present study from the harder magnetization components 5 and 6 (Table 1, Fig. 7F), agrees very well with the palaeoposition of the Indus-Tsangpo suture at about 50 m.y. B.P., as deduced from the DSDP core data (Fig. 7B). This observation clearly supports collision of the Indian subcontinent with the mass of the Ladakh island arc, at pre-Middle Eocene times. This collision had occurred most probably already by Late Palaeocene times as indicated by the timing of emplacement of ultramafic complexes in eastern Baluchistan and the findings in this region of collision related secondary magnetization components for which a Late Palaeocene age of acquisition can be inferred [81].

As shown from the DSDP core data (Fig. 7B, C), the Indian subcontinent continued its northward movement between about 50 m.y. and about 40 m.y., and at a slower rate thereafter. Comparison with the palaeomagnetic data from northern Eurasia suggest intimate contact between India and Asia at about 40 m.y. (Fig. 7C).

The palaeolatitude of the Ladakh Intrusives, according to component 4 (Figs. 6, 7F) agrees rather well with the palaeomagnetically determined palaeopositions of both the Indus-Tsangpo suture zone (DSDP core data) and the suggested southern boundary of Asia as the eastwards continuation of the Zagros-Makran zone [55,74]. This suggests that acquisition of this component may be related to final collision between India and Asia.

Between initial collision of the Indian subcontinent with the postulated Ladakh island arc (about 53 m.y. ago according to Powell [55]) and final intimate collision with southern Eurasia (about 38 m.y. ago according to Molnar and Tapponnier [14]), the Indian subcontinent together with the Ladakh island arc welded to it, moved northward over more than 10° of latitude (Fig. 7F). Frank et al. [6] have suggested that consumption of the former crust between the Indian subcontinent and southern Asia did not take place along one single subduction zone. It may well be that the major change in northward movement at about the Eocene/Oligocene time boundary occurred when earlier and more northerly situated subduction zones, such as the postulated Shyok zone [6] to the north of the Ladakh Intrusives, became inactive.

Subsequent to final collision, the Indian subcontinent has indented over more than 10° of latitude into south-central Asia (Fig. 7). This palaeomagnetic observation substantiates preliminary palaeomagnetic data from the Chitral area [42] situated immediately north of the possible westward continuation of the Indus-Tsangpo suture zone, and also from eastern Baluchistan [81], which suggest crustal shortening in south-central Asia over more than 20° of latitude, and is

well in accord with preliminary palaeomagnetic data from south-central Tiber [43,44]. Our palaeomagnetic data generally support the suite of models envisaging indentation of Asia by the northwards-moving Indian continental fragment (cf. [8,13–17,18,30,55]), and are well in accord with Molnar and Tapponnier's [14] interpretation of ERTS photographs that more than 1000 km of crustal shortening in south-central Asia must have occurred since the India-Asia collision.

It is interesting to note that the older and magnetically "harder" components 5 and 6 have a northwards declination, whereas the younger and magnetically "softer" components 3 and 4 have a north-northeastwards declination. It is possible that the eastwards-directed and magnetically soft component 2 has not been separated completely from components 3 and 4 because of an unnoticed composite breakdown. In an alternative and more likely interpretation, however, this declination sequence suggests that during and after initial collision of the Indian subcontinent with the postulated Ladakh island arc, the sampled region underwent a slight counterclockwise rotation (10–20° or more) due to drag of the counterclockwise-rotating Indian plate (fig. 7). This movement reversed to a 10–20° clockwise rotation after final collision with south-central Asia when the sampled region became welded to the clockwise-rotating Eurasian plate.

Various models for the original extent of the Indian fragment of Gondwanaland have been summarized here and before [42]. The palaeomagnetic results presented from the Ladakh Intrusives represent palaeomagnetic directions which originated after initial collision with the Indian subcontinent. Although the data obtained place additional constraints on the various models for the original extent of Gondwanic India, they are not directly relevant to test hypotheses that the Tibetan Plateau [28,29] or parts of it [31,34] may have originally formed part of Gondwanaland. Powell and Conaghan's model [8,18,30,55] of "Greater India's" wedge-shaped underthrusting of the Tibetan Plateau predicts large-scale underthrusting in the Ladakh region. The present data accord with the latter aspect as post-collisional northward movement of the Indian subcontinent, as deduced from sea-floor spreading data, is comparable in magnitude to the here palaeomagnetically detailed northward movement of the collision zone in the Ladakh region. Large-scale underthrusting in the east, as predicted in their model, still has to be tested, however.

Further work may show the here advanced two-stage collisional model for the Ladakh area to be representative also for the wider Himalayan region. If so, it can be interpreted in support of one of Bird's [83] models for the origin of the middle to late Tertiary anatectic leucogranites north of the MCT zone, as resulting from high-level infilling of hot astenospheric material into a delaminated lithosphere. In Bird's model III a delamination of the lithosphere is hypothized through backwards-bending of the subducting slab of the northwards moving Indian plate. This slab is subducting underneath an also northwards-moving overriding unit, casu an island arc off south-central Asia.

7. Conclusions

Six different magnetization components have been identified in the present sample collection from the Ladakh Intrusives at Kargil.

A northwards- and downwards-directed component 1 and another eastwards- and slightly downwards-directed component 2 are present throughout the sample collection. Both components are of a magnetically "soft" character, are of recent origin, have originated through low-temperature oxidation, and represent the effect of respectively (1) the present local field at the sampled area, and (2) a grossly east-west-aligned foliation pattern related to collision tectonics.

The other four magnetization components (3, 4, 5 and 6) originated during various stages of the India-Asia collision. The two magnetically "softer" components (3 and 4, group B) indicate a palaeoposition of the sampled area to the south of its present position. Component 4 indicates crustal shortening in southern Asia over more than 10° of latitude, which may well be related to final collision of the Indian subcontinent/Ladakh island arc mass with south-central Asia (about 38 m.y. ago, according to Molnar and Tapponier [14]). Component 3 indicates a palaeoposition of the sampled area much nearer to its present position and may well have originated during the younger phases of the Himalayan Orogeny (of Mio-Pliocene and younger age).

The two magnetically "harder" components (5 and

6, group A) have originated during cooling shortly after initial collision of the Indian subcontinent with the Ladakh island arc (at about 53 m.y. ago, according to Powell [55]). The palaeoposition of the sampled area as determined from components 5 and 6 is in very good agreement with the palaeoposition of the Indus-Tsangpo suture zone as deduced from DSDP core data from the Indian plate, and indicates that intial collision of the Indian subcontinent with the Ladakh island arc took place at a latitude of about 7–10°N, or possibly at even farther southern latitudes. This is far to the south of its present position at about 34°N in the Ladakh area.

After collision of the Indian subcontinent along the Indus-Tsangpo suture zone with the Ladakh island arc, this composite unit moved northwards over more than 10° of latitude before final collision with southern Asia occurred along a farther northwards and presently inactive suture zone, possibly represented by the Shyok zone to the north of the Ladakh Range [6].

The older magnetization components 5 and 6 have northwards declinations, whereas the younger magnetization components 3 and 4 have north-northeast declinations. This suggests counterclockwise rotation of the sampled area after initial collision of the counterclockwise-rotating Indian subcontinent with the Ladakh island arc, followed by a clockwise rotation of the sampled area after final collision of this now combined unit with the clockwise-rotating Asian plate.

The present palaeomagnetic data indicate substantial crustal shortening in south-central Asia after collision. This is well in agreement with the interpretation of preliminary palaeomagnetic data from Chirtral [42] and southcentral Tibet [43,44], and supports crustal shortening in south-central Asia over more than 1000 km as postulated by Molnar and Tapponnier [14] from studies of ERTS photographs.

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